

Discretization by Classification & Correlation Analysis

□ Classification (e.g., decision tree analysis)

- Supervised: Given class labels, e.g., cancerous vs. benign
- Using *entropy* to determine split point (discretization point)
- Top-down, recursive split
- Details to be covered in Chapter “Classification”

□ Correlation analysis (e.g., Chi-merge: χ^2 -based discretization)

- Supervised: use class information
- Bottom-up merge: Find the best neighboring intervals (those having similar distributions of classes, i.e., low χ^2 values) to merge
- Merge performed recursively, until a predefined stopping condition